

# GPRB Wireless PSI Technical MCQ Practice

Initial PDF set for revision. Based on official GPRB/OJAS Advt. GPRB/202526/2 exam pattern.

## Exam Pattern Snapshot

| Part   | Subject                                                                | Questions | Marks |
|--------|------------------------------------------------------------------------|-----------|-------|
| Part A | Reasoning, Quantitative Aptitude, Constitution, Current Affairs and GK | 80        | 80    |
| Part B | Technical Subject                                                      | 120       | 120   |
| Total  | Objective MCQ exam                                                     | 200       | 200   |

Time: 3 hours. Each question: 1 mark. Wrong attempted answer: -0.25. Blank question without selecting any option: -0.25. Use option E / Not Attempted when unsure.

## Technical Syllabus List

- 1 Electronics Components, Devices and Circuits
- 2 Digital Electronics and VLSI
- 3 Electronics Networks and Instruments and Measurements
- 4 Communication Engineering
- 5 Communication Applications
- 6 Microprocessors and Microcontrollers
- 7 Computer Networks
- 8 Essentials of Network Security
- 9 Web Technology
- 10 Android Application Development
- 11 Current Trends: 4G LTE, POLNET, disaster communication, real-time surveillance, recent advancements

## Practice MCQs

**1. Intrinsic semiconductor means:**

- A. A pure semiconductor without intentional impurity
- B. A conductor with high impurity
- C. An insulator with zero conductivity
- D. A semiconductor only with pentavalent impurity

**2. Silicon and germanium are examples of:**

- A. Conductors
- B. Semiconductors
- C. Insulators only
- D. Magnetic materials

**3. In an N-type semiconductor, majority carriers are:**

- A. Holes
- B. Electrons
- C. Neutrons
- D. Positive ions only

**4. N-type semiconductor is formed by adding:**

- A. Trivalent impurity
- B. Pentavalent impurity
- C. Divalent impurity
- D. Monovalent impurity

**5. P-type semiconductor is formed by adding:**

- A. Trivalent impurity
- B. Pentavalent impurity
- C. No impurity
- D. Only silicon dioxide

**6. Majority carriers in P-type semiconductor are:**

- A. Electrons
- B. Holes
- C. Photons
- D. Negative ions

**7. The barrier potential of a silicon diode is approximately:**

- A. 0.1 V
- B. 0.3 V
- C. 0.7 V
- D. 1.5 V

**8. The barrier potential of a germanium diode is approximately:**

- A. 0.3 V
- B. 0.7 V
- C. 1.2 V
- D. 5 V

**9. A PN junction diode conducts current mainly in:**

- A. Reverse bias
- B. Forward bias
- C. Zero bias only
- D. Open circuit condition only

**10. In reverse bias, diode current is ideally:**

- A. Very high
- B. Zero

- C. Equal to forward current
- D. Only AC current

**11. Depletion region in a PN junction contains:**

- A. Free electrons and holes in large quantity
- B. Immobile ions
- C. Only photons
- D. Only protons moving freely

**12. Reverse saturation current in a diode is mainly due to:**

- A. Majority carriers
- B. Minority carriers
- C. External resistor
- D. Battery polarity only

**13. Zener diode is normally operated in:**

- A. Forward bias only
- B. Reverse breakdown region
- C. Open circuit region
- D. Saturation region of BJT

**14. Main use of Zener diode is:**

- A. Voltage regulation
- B. Current amplification
- C. Radio transmission
- D. Magnetic storage

**15. LED emits light when it is:**

- A. Forward biased
- B. Reverse biased beyond breakdown
- C. Connected as an open circuit
- D. Used as a capacitor

**16. A photodiode is generally operated in:**

- A. Forward bias
- B. Reverse bias
- C. Saturation of transistor
- D. Short circuit only

**17. Rectifier is used to convert:**

- A. DC to AC
- B. AC to DC
- C. Low frequency to high frequency only
- D. Analog signal to digital signal only

**18. Ripple factor is related to:**

- A. Unwanted AC component in rectifier output
- B. Magnetic flux density
- C. Only antenna gain
- D. Frequency modulation index

**19. A filter circuit in a power supply is used to:**

- A. Increase ripple
- B. Reduce ripple
- C. Convert DC to RF
- D. Destroy the diode

**20. The peak inverse voltage is important for selecting:**

- A. Diode in rectifier circuit
- B. Only resistor color code
- C. Only capacitor dielectric color
- D. Logic gate family

**21. BJT stands for:**

- A. Base Junction Transmitter
- B. Bipolar Junction Transistor
- C. Binary Junction Transformer
- D. Balanced Junction Terminal

**22. A BJT has how many terminals?**

- A. 2
- B. 3
- C. 4
- D. 5

**23. The three terminals of a BJT are:**

- A. Anode, cathode, gate
- B. Emitter, base, collector
- C. Source, gate, drain
- D. Input, output, ground only

**24. In normal active mode of NPN transistor:**

- A. Emitter-base junction is forward biased and collector-base junction is reverse biased
- B. Both junctions are reverse biased
- C. Both junctions are forward biased
- D. Emitter-base junction is reverse biased and collector-base junction is forward biased

**25. Common emitter amplifier provides:**

- A. Voltage gain and phase reversal
- B. No voltage gain
- C. Only impedance matching with gain less than 1 always
- D. Only rectification

**26. Current gain in common emitter configuration is denoted by:**

- A. Alpha
- B. Beta
- C. Gamma ray
- D. Omega

**27. Transistor biasing is required to:**

- A. Set proper operating point
- B. Remove all DC supply
- C. Make transistor an insulator
- D. Convert it into a diode only

**28. Thermal runaway in transistor is related to:**

- A. Decrease in temperature only
- B. Increase in collector current due to temperature rise
- C. Only antenna radiation
- D. Only digital clock pulse

**29. A transistor can be used as:**

- A. Amplifier and switch
- B. Only fuse
- C. Only transformer
- D. Only mechanical relay

**30. In cutoff region, a transistor acts approximately as:**

- A. Closed switch
- B. Open switch
- C. Perfect battery
- D. Capacitor only

**31. Binary number system has base:**

- A. 2
- B. 8
- C. 10
- D. 16

**32. Decimal number 10 is equal to which binary number?**

- A. 1010
- B. 1001
- C. 1110
- D. 0101

**33. The output of AND gate is 1 when:**

- A. All inputs are 1
- B. Any one input is 0
- C. All inputs are 0
- D. Inputs are floating

**34. The output of OR gate is 0 when:**

- A. All inputs are 0
- B. Any one input is 1
- C. All inputs are 1
- D. One input is high

**35. NOT gate is also called:**

- A. Buffer
- B. Inverter
- C. Adder
- D. Counter

**36. Universal gates are:**

- A. AND and OR
- B. NAND and NOR
- C. XOR and XNOR only
- D. NOT and buffer

**37. Flip-flop is used to store:**

- A. One bit of information
- B. One byte always
- C. Only analog voltage continuously
- D. Only RF signal

**38. A counter is made using:**

- A. Flip-flops
- B. Only resistors
- C. Only transformers
- D. Only antennas

**39. ADC converts:**

- A. Analog signal to digital signal
- B. Digital signal to analog signal
- C. AC to DC only
- D. DC to AC only

**40. DAC converts:**

- A. Analog signal to digital signal
- B. Digital signal to analog signal
- C. High voltage to zero voltage only
- D. Optical signal to RF only

## Answer Key

1-A, 2-B, 3-B, 4-B, 5-A

6-B, 7-C, 8-A, 9-B, 10-B

11-B, 12-B, 13-B, 14-A, 15-A

16-B, 17-B, 18-A, 19-B, 20-A

21-B, 22-B, 23-B, 24-A, 25-A

26-B, 27-A, 28-B, 29-A, 30-B

31-A, 32-A, 33-A, 34-A, 35-B

36-B, 37-A, 38-A, 39-A, 40-B

## How To Use This PDF

- 1 First attempt questions without seeing answer key.
- 2 Mark weak areas: diode, transistor, digital basics, etc.
- 3 Revise formulas and definitions for wrong answers.
- 4 Re-attempt the same set after 2 days.